



Geometry introduces students to spatial thinking. Students learn to dissect complex geometrical configurations and use prior results to support their claims. This geometry course acts as an introduction to formal proof techniques, helping students transition to a more mature view of mathematics. Geometry problems are also tackled analytically by placing them on the Cartesian plane. In addition, we introduce students to trigonometry to build their foundation for precalculus and calculus. The challenging problems in this course are all contest-level problems, including real questions from MATHCOUNTS, AMC 10, AMC 12, and other prestigious competitions.

Book: *Introduction to Geometry*, by Richard Rusczyk

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Content:

Week	Topic	Week	Topic
1	Angles, Parallel Lines, & Proofs	19	Geometric Inequalities II
2	Angles in a Triangle	20	Sectors & Circular Segments
3	Congruent Triangles	21	Circles: Angles
4	Isosceles & Equilateral Triangles	22	Circles: Tangent Lines
5	Perimeter & Area	23	Circles: Power of a Point
6	Similar Triangles I	24	Midterm 2
7	Similar Triangles II	25	3D Geometry I
8	Right Triangles	26	3D Geometry II
9	Perpendicular Bisectors and Altitudes	27	3D Geometry III
10	Concurrency and Angle Bisectors	28	Geometric Transformations I
11	Medians and Ceva's Theorem	29	Geometric Transformations II
12	Midterm 1	30	Analytic Geometry I
13	Quadrilaterals: Trapezoids	31	Analytic Geometry II
14	Quadrilaterals: Rhombi & Kites	32	Analytic Geometry III
15	Quadrilaterals: Rectangles & Squares	33	Trig: The Basics
16	Polygons I	34	Trig: Law of Sines and Cosines
17	Polygons II	35	Trig: Identities
18	Geometric Inequalities I	36	Final Exam

Common Core State Standards:

Domain	Subdomain	Standards
Algebra	Reasoning with Equations and Inequalities	7
Functions	Trigonometric Functions	3
Geometry	Congruence	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13
	Similarity, Right Triangles, & Trigonometry	1ab, 2, 3, 4, 5, 6, 7, 8
	Circles	1, 2, 3, 4, 5
	Expressing Geometric Properties with Equations	1, 4, 5, 6, 7
	Geometric Measurement & Dimension	1, 2, 3, 4
	Modeling with Geometry	3

Sample Problems:

- It is possible for two angle bisectors in a triangle to be perpendicular? Why or why not?
- Point N is on hypotenuse $\overline{BC}$ of right triangle $\triangle ABC$ such that $\angle CAN = 45^\circ$. Given $AC = 8$ and $AB = 6$, find AN .

► Equilateral triangle ABC is inscribed in a circle. Let P be a point on arc BC , and let $\overline{AP}$ intersect $\overline{BC}$ at Q . Prove that

$$\frac{1}{PQ} = \frac{1}{PB} + \frac{1}{PC}.$$

- Point X is on diameter $\overline{PQ}$ of a circle. Prove that if $\overline{AB}$ is a chord of the circle parallel to $\overline{PQ}$, then $XA^2 + XB^2 = XP^2 + XQ^2$.
- Let $ABCD$ be a square, and let P be a point on side $\overline{CD}$. Construct squares $BPWX$ and $APYZ$ externally on $\overline{BP}$ and $\overline{AP}$, respectively, as shown. Let O, O_1 , and O_2 be the centers of squares $ABCD, BPWX$, and $APYZ$, respectively. Show that quadrilateral OO_1PO_2 is a parallelogram.

